

and the correct course is unclear. Sometimes, our decisions have significant consequences. These situations demand substantial time and effort to try to devise a systematic approach to analyzing various courses of action.

Even in a perfect world, when a decision maker must choose one among a number of possible actions, the ultimate consequences of most, if not every, available action will depend on uncertainties to be resolved in the future. When deciding under uncertainty, there are generally accepted guidelines that a decision maker should follow:

1. Take an inventory of all viable options available for obtaining information, for experimentation, and for action.
2. List the events that may occur.
3. Arrange pertinent information and choices/assumptions.
4. Rank the consequences resulting from the various courses of action.
5. Determine the probability of an uncertain event occurring.

Unfortunately, facing uncertainty, most people cannot and do not systematically describe problems, record all the necessary data, or synthesize information to create rules for making decisions. Instead, most people venture down somewhat more subjective, less ideal paths of reasoning in an attempt to determine the course of action consistent with their basic judgments and preferences. How, then, can decision making be faithfully modeled?

Raiffa In 1968, in *Decision Analysis: Introductory Lectures on Choices under Uncertainty*,⁹ decision theorist Howard Raiffa introduced to the analysis of decisions three approaches that provide a more accurate view of a “real” person’s thought process:

1. *Normative* analysis is concerned with the rational solution to the problem at hand. It defines an ideal that actual decisions should strive to approximate.
2. *Descriptive* analysis is concerned with the manner in which real people actually make decisions.
3. *Prescriptive* analysis is concerned with practical advice and tools that might help people achieve results more closely approximating those of normative analysis.

Raiffa’s contribution laid the foundation for a significant work in the field of BFMI, an article by Daniel Kahneman and Mark Riepe entitled “Aspects of Investor Psychology: Beliefs, Preferences, and Biases Investment Advisors Should Know About.” This work was the first to tie together decision theory and financial advising. According to Kahneman and Riepe,